Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	14 October 2026
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

The following functional requirements define the core behaviour of the proposed ServiceNow-based Automated Network Request Management solution. They are derived from the project's Service Catalog journey, the documented Workflow Studio implementation, approval path, record updates and notification actions. Where a capability is not visible in the supplied implementation evidence, it is treated as planned or future scope rather than claimed as implemented.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Network Request Catalog	Provide a Service Catalog item for employees/users to submit network-related requests. Acceptance: Capture the configured request information and create a ServiceNow Requested Item (RITM) from the submission.
FR-2	Request Data Capture	Capture required request information through catalog variables and mandatory fields. Acceptance: Configured variables are available to the workflow; incomplete required information cannot be submitted.
FR-3	Dynamic Form Behaviour	Use clear form behaviour to show or require fields according to the request context. Acceptance: Conditional visibility and mandatory behaviour work as configured without unnecessary client-side logic.
FR-4	Request Submission & Workflow	Trigger the Network Request Flow Designer process when the catalog request is submitted. Acceptance: The Service Catalog trigger receives the Requested Item Record and starts the configured flow.
FR-5	Catalog Variable Retrieval	Retrieve selected variables from the Network Request catalog item for downstream automation. Acceptance: The Get Catalog Variables action exposes the configured variables as flow outputs.
FR-6	Operational Record Creation	Create a structured operational record from the submitted request values. Acceptance: Configured catalog values are mapped into the operational record and Approval Status is initialized as Requested.
FR-7	Approval Management	Send the created network request record through the configured approval action. Acceptance: An approval request is generated and the configured approver/approval rule can provide an outcome.
FR-8	Approval Decision & Branching	Evaluate the approval outcome and execute the correct workflow path. Acceptance: Approved requests follow the Approved branch; non-approved requests follow the Else path and receive the corresponding record update.
FR-9	Request Notifications	Send lifecycle notifications at configured request stages. Acceptance: The request-created notification is sent after record creation and the configured resolution notification is sent after the approved update.
FR-10	Request Status & Tracking	Maintain an understandable request status across the workflow. Acceptance: The operational record reflects the configured lifecycle state such as Requested, Approved or Rejected.
FR-11	Fulfilment Handoff & Operations	Make the approved request context available for downstream network fulfilment and operational tracking. Acceptance: The approved request contains the mapped request information needed by the responsible operational process; detailed fulfilment automation is a future extension unless configured.
FR-12	Workflow Monitoring & Handoff Documentation	Support Flow Designer execution review, troubleshooting and solution handoff. Acceptance: Administrators can inspect execution details, mappings and condition paths and maintain documentation for recreation in another ServiceNow instance.

Non-functional Requirements:

The non-functional requirements define the quality expectations for usability, security, reliability, performance, availability, scalability, maintainability, data integrity, compatibility and auditability. These requirements are expressed at the application-design level; platform-level guarantees remain dependent on the deployed ServiceNow instance.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The requester-facing catalog experience should be simple, readable and guided. Mandatory fields and clear descriptions should reduce incomplete submissions.
NFR-2	Security	Access to network-request and operational data should be governed through ServiceNow roles and ACLs, with administrative configuration restricted to authorized users.
NFR-3	Reliability	For valid submissions, catalog data should move consistently from the Service Catalog through Flow Designer into the configured operational record.
NFR-4	Performance	Normal request submission and workflow processing should avoid unnecessary manual steps and excessive client-side processing; server-side orchestration should remain maintainable.
NFR-5	Availability	The solution is expected to be accessible whenever the deployed ServiceNow instance and required platform services are available; no independent uptime guarantee is asserted by this project.
NFR-6	Scalability	The design should allow additional network request categories, variables, approval rules, fulfilment paths and notification rules without redesigning the complete solution.
NFR-7	Maintainability	Prefer Service Catalog, Flow Designer, UI Policies and ACL configuration. Custom scripting should be limited to necessary behaviour and the workflow should remain understandable to another administrator.
NFR-8	Data Integrity	Values captured in the catalog should remain consistent when mapped to the operational record. Mapping and validation errors should be detectable during testing.
NFR-9	Compatibility	The requester experience should operate in the intended ServiceNow Service Portal/catalog context and use supported platform capabilities for workflow and notifications.
NFR-10	Auditability & Troubleshooting	Administrators should be able to inspect workflow execution, approval/condition paths, stored values and available execution information to diagnose failures.

Future / Out-of-Scope Requirements for the Current Version

• Additional network request categories and specialized catalog items beyond the currently demonstrated Network Request item.

• Multi-level approval chains such as manager approval followed by network-security approval, unless configured in the implementation.

• Automated network-device/API integrations for direct configuration changes.

• Automated fulfilment task creation and closure rules beyond the workflow currently demonstrated.

• Advanced SLA calculation, escalation and performance dashboards.

• External messaging channels such as SMS, WhatsApp or other notification integrations.

• AI-based request classification, priority prediction or anomaly detection.

• Advanced network analytics and operational dashboards.

Requirement Traceability Summary

The critical end-to-end requirement path is: requester enters valid network-request information → request is submitted through the Service Catalog → Flow Designer retrieves catalog variables → the operational record is

created with an initial Requested state → approval is requested → the approval result is evaluated → the record is updated to Approved or Rejected → the configured notification is sent → the resulting request is available for operational fulfilment and tracking. This path defines the highest-priority testing and troubleshooting scenarios, especially variable mappings, record creation, approval routing, condition logic and notification execution.